Exam 2 Study Guide

STAT 251 SECTION 01

2026-10-12

Exam Logistics:

- Exam 2 is a **take-home, open-note exam** completed in **Canvas**. You may use your notes, the textbook and the course website. It is due **Sunday, October 18, 2026 at 11:59PM**. An in-class review will be held on **Wednesday, October 14, 2026**
- The exam is 25 questions plus 2 bonus questions. Most are multiple choice or true/false, and four are short written explanations

How to prepare

The exam asks you to **interpret** values, **identify** the correct equation, and **choose** the right method for a situation. Some questions are **"show how to compute"** problems: a short story problem gives you every number you need, and you choose the formula with the correct values filled in. You will not have to carry out long calculations or look up values in a table, but you do need to recognise which formula fits, which numbers go where, and the common mistakes (for example, using σ where $\sigma/\sqrt{n}$ belongs, or forgetting to take the complement for a "greater than" probability).

Because the exam is open-note, there are no review problems for Exam 2. Use the list below to check that your notes cover everything, and make sure you can explain each item in your own words. Homework 2 (../homeworks/2026/homework_2_2026.html) covers most of this material, and some exam problems are based on homework problems.

Content To Study

Exam 2 will cover material beginning on **Monday of Week 4 (September 14)** through **Monday of Week 8 (October 12)**. This includes the following topics:

- Sampling designs, bias, sampling and nonsampling error - September 14 and 21
- Experimental design, association vs. causation and research ethics - September 23 and 25
- The normal distribution, z-scores, the empirical rule and identifying outliers - September 25, 28 and 30

- Normal quantiles (z-table, `pnorm` and `qnorm`) and transformations of variables - September 30
- Standard error, margin of error and statistical significance - September 30 and October 2
- Probability theory, random variables and discrete distributions - October 2, 5, 7 and 9
- Continuous distributions (through the normal distribution) - October 9
- Sampling distributions and the Central Limit Theorem - October 12
- Introduction to confidence intervals - October 12

Sampling Designs

1. Know the terminology for sampling designs: **population**, **sample**, **sampling frame**, **census** and **sample survey**
2. Know how to define and identify the sampling designs we talked about in class: **simple random sampling**, **systematic sampling**, **stratified random sampling**, **cluster sampling** (including **two-stage cluster sampling**), **convenience sampling** and **voluntary response sampling**
3. Know the difference between a **stratum** (members alike within, and a sample is taken from *every* stratum) and a **cluster** (a mini-population, where only *some* clusters are selected and everyone inside them is measured)
4. Know the difference between a convenience sample and a voluntary response sample (who does the choosing), and why neither one is a good way to sample a population
5. Know the difference between **sampling with replacement** and **sampling without replacement**
6. Know the difference between **sampling error** and **nonsampling error**, and what makes a sample **biased** or **representative** (textbook Sections 1.2 and 1.4)
7. Know the sources of bias in a sample survey: **undercoverage**, **nonresponse** and **response bias**
8. Know why a **larger sample reduces sampling error but does not remove bias**: a bigger biased sample gives a more precise wrong answer

Experimental Design and Ethics

1. Know the definitions of an **explanatory variable**, a **response variable**, an **experimental study**, an **observational study** and a **survey**, and be able to explain the difference between an experiment and an observational study
2. Know the language of experiments: **experimental units / subjects**, **factors**, **levels**, **treatments**, **control group**, **placebo**, **blinding** and **double blinding**
3. Know how to define and identify the three experimental designs we discussed: **completely randomized design**, **randomized block design** (and what a **blocking variable** is) and **matched pairs design**
4. Know the difference between **statistical association** and **causation**, and what a **lurking** (confounding) variable is. Be able to explain why an observational study on its own cannot show that one variable causes another
5. Know what **random sampling** (generalizing to a population) and **random assignment** (concluding cause) each allow you to conclude
6. Know the basic ethical requirements for research on human subjects: **informed consent**, **confidentiality** and **IRB** review
7. Recognize unethical data practices, such as fabricating data or quietly leaving out people who were hard to reach (textbook Section 1.4)

The Normal Distribution

1. Know what a normal distribution looks like (a continuous, symmetric, bell-shaped curve), and that the total area under the curve is 1
2. Know how the mean μ (location) and the standard deviation σ (spread) affect a normal curve, and how two normal curves compare when they share a mean or share a standard deviation
3. Know the **empirical rule** (approximately 68%, 95% and 99.7% of observations within 1, 2 and 3 standard deviations of the mean), and how to use it together with symmetry to find the percentage above or below a value, or the cutoff for the top 2.5%
4. Know how to compute and interpret a z-score, read in **standard deviations** above or below the mean, and how it relates to the **standard normal distribution**. Know how to use z-scores to compare observations from two different distributions
5. Know how to write a normal probability as a standard normal probability: a Z-table gives the area to the **left**, $P(Z \leq z)$, so a "greater than" probability needs the complement and a "between" probability is a difference of two left areas

6. Know how to identify outliers with the $\pm 2s$ **rule** and with the $1.5 \times IQR$ **rule**, how to judge from a boxplot whether a distribution looks roughly normal, and why the $\pm 2s$ rule should only be trusted for roughly normal data. Know why the two rules can disagree: one extreme value inflates s but barely moves the quartiles
7. Know how to find an area for a z-score that is not a whole number of standard deviations, from a z-table or with `pnorm`, and how to go from an area back to a value with `qnorm`

Transformations of Variables

1. Know the difference between a **linear transformation**, $y = ax + b$ (adding, subtracting, multiplying, dividing, such as changing units), and a **nonlinear transformation** (squaring, roots, logarithms)
2. Know what a linear transformation does to the summaries: the center moves with the whole transformation ($\bar{y} = a\bar{x} + b$, and the same for the median), the spread is multiplied by $|a|$ and ignores the shift ($s_y = |a|\, s_x$, $IQR_y = |a|\, IQR_x$), and the **shape does not change**
3. Know that a z-score is a linear transformation, so standardizing skewed data does **not** make it normal

Probability Theory

1. Be comfortable with the language of probability theory: know how to define **probability**, **sample space**, **event**, **random trial**, and the **union** and **intersection** of two events
2. Know the rules we discussed for probability: the **complement rule**, the **addition rule** and the **multiplication rule**, and when each one applies (“not”, “or”, “and”)
3. Know how to compute the probability of a **complement** of an event, the **union** of two events (subtracting the overlap so it is not counted twice), and the **intersection** of two independent events
4. Know how to find an “at least one” probability through its complement, “none”
5. Know how to compute probabilities when sampling with vs. without replacement: without replacement, the second probability is computed from what is left after the first draw
6. Know how to use a Venn diagram to show the relationships between events and fill in the count for each region

7. Know what is meant when two events are **disjoint (mutually exclusive)** or **independent**, and why disjoint events are not independent
8. Know what is implied by the **Law of Large Numbers** (the relative frequency of an event approaches its true probability as the number of observations increases), and why it does not mean a result is "due" to happen (the gambler's fallacy)

Random Variables and Discrete Distributions

1. Know the definitions of a **random variable**, and the difference between **discrete** and **continuous** random variables
2. Understand what is meant by the **probability distribution** of a discrete random variable, and how to check that a table of probabilities is a valid distribution (each probability between 0 and 1, and the probabilities add to 1)
3. Be able to compute the **mean**, **variance** and **standard deviation** of a discrete random variable from its probability distribution. The mean is a **probability-weighted** average, $\mu = \sum_x x \cdot P(x)$, not the plain average of the possible values
4. Know the **Bernoulli** distribution (one trial, success with probability p), and that its mean is p and its variance is $p(1-p)$
5. Know when a count follows a **binomial** distribution (a fixed number n of independent trials, each a success with the same probability p), how to check all three conditions (for example, drawing without replacement from a small population breaks independence), how to compute a probability with its probability mass function, what the $\frac{n!}{k!(n-k)!}$ term counts, and that its mean is np and its variance is $np(1-p)$
6. Know when a count follows a **Poisson** distribution (events occurring independently at a constant average rate λ in a fixed interval of time or space), how to compute a probability with its probability mass function, that its mean and variance are both λ, and how to rescale λ when the interval changes
7. Know how to compute "at most" and "at least" probabilities, such as $P(X \leq 1) = P(X = 0) + P(X = 1)$ and $P(X \geq 2) = 1 - P(X \leq 1)$, from the probability mass function

Continuous Distributions

1. Know the difference between **probability mass** (a discrete variable: each value carries its own probability) and **probability density** (a continuous variable: probability is the **area under the density curve**, and the height of the curve is not a probability)
2. Know the properties of a density curve: it is never negative and the total area under it is 1. Know why $P(X = c) = 0$ for a continuous random variable, so $P(X < c) = P(X \leq c)$
3. Know the definition of the **cumulative distribution**, $P(X \leq x)$ (the area to the left, which is what the z-table and `pnorm` give), and how to get the area to the right, $P(X > x) = 1 - P(X \leq x)$, and the area between, $P(a < X < b) = P(X \leq b) - P(X \leq a)$
4. Know that the quartiles of a continuous distribution cut the area under the curve into quarters, and that the mean is the balance point of the curve

Sampling Distributions and Statistical Significance

1. Know the three distributions and keep them apart: the **population distribution** (how a single observation varies; described by parameters such as μ, σ and p), the **data distribution** (the n values in your one sample; described by statistics such as $\bar{x}$, s and $\hat{p}$) and the **sampling distribution** (how a statistic such as $\bar{x}$ varies over all possible samples of the same size n)
2. Know why two random samples from the same population give different statistics (**sampling variability**), and that a statistic is a random variable with its own distribution
3. Know how to build the sampling distribution of $\bar{x}$ for a very small population and sample size: list every possible sample (order matters) and its mean, find each sample's probability (multiply, for independent observations), and add up the samples that give the same value of $\bar{x}$
4. Know the center and spread of the sampling distribution of $\bar{x}$: its mean is μ ($\bar{x}$ is **unbiased**), and its **standard error** is $\sigma_{\bar{x}} = \sigma/\sqrt{n}$. Know why it is smaller than σ (σ describes single observations, the standard error describes whole-sample averages) and how it changes with the sample size (making the sample 4 times larger cuts the standard error in half)
5. Know the mean and standard error of the sampling distribution of the sample proportion $\hat{p}$: mean p and standard error $\sqrt{p(1-p)/n}$
6. Know the **margin of error** as $2 \times SE$, how it changes as n grows, and the difference between the margin of error (a bound we can compute in advance) and the sampling error (how far this particular estimate is from the parameter, which we never know)

7. Know what makes a result **statistically significant**: it falls more than 2 standard errors from what chance alone would give, that is, outside the margin of error. Know that significant does not mean important, and not significant does not mean no difference

The Central Limit Theorem

1. Understand the **Central Limit Theorem**: for large n, the sampling distribution of $\bar{x}$ is approximately normal, $N(\mu, \sigma/\sqrt{n})$, **whatever the shape of the population**. As n grows it becomes more bell shaped and narrower; the population distribution itself does not change
2. Know the rules of thumb for "large enough": $n \geq 30$ for $\bar{x}$, and $np > 5$ and $n(1-p) > 5$ for $\hat{p}$
3. Know how to use the CLT: about 95% of samples give a statistic within 2 standard errors of its mean, and any other probability about $\bar{x}$ comes from the z-score $z = (\bar{x} - \mu)/(\sigma/\sqrt{n})$

Introduction to Confidence Intervals

1. Know the difference between a **point estimate** (a single statistic used to estimate a parameter: $\bar{x}$ for μ, $\hat{p}$ for p) and an **interval estimate**
2. Know that a **confidence interval** has the form point estimate $\pm$ margin of error, and that $\bar{x} \pm 2\sigma/\sqrt{n}$ is a 95% confidence interval for μ (it comes from turning "$\bar{x}$ is within 2 standard errors of μ" around)
3. Know how to report an interval: "We estimate with 95% confidence that the population mean is between [lower bound] and [upper bound]"
4. Know what "95% confident" means: about 95% of the intervals built this way capture the parameter. The parameter is fixed, so any one interval either does or does not contain it. Be able to recognize the two common wrong readings: "there is a 95% probability that μ is in this interval" and "95% of the individual values fall in this interval"

Formulas:

- Be sure you understand the notation (i.e., symbols) we have used so far (e.g., $n, N, s, s^2, \bar{x}, \mu, \sigma, \sigma^2, \hat{p}, p, \lambda$)

The following formulas will be provided on the exam, **unlabeled** – the equations only, with no names or descriptions. Part of what the exam asks is which formula fits the situation in front of you.

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i, \quad s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2, \quad s = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

$$z = \frac{x - \mu}{\sigma}, \quad P(X > x) = 1 - P(X \le x), \quad P(a \le X \le b) = P(X \le b) - P(X \le a)$$

$$\mu = \sum_x x \cdot P(x), \quad \sigma^2 = \sum_x (x - \mu)^2 \cdot P(x), \quad \sigma = \sqrt{\sum_x (x - \mu)^2 \cdot P(x)}$$

$$P(A') = 1 - P(A), \quad P(A \cup B) = P(A) + P(B) - P(A \cap B), \quad P(A \cap B) = P(A) \cdot P(B)$$

$$P(X = k) = \frac{n!}{k!(n-k)!} \cdot p^k \cdot (1-p)^{n-k}, \quad P(X = k) = \frac{\lambda^k \cdot e^{-\lambda}}{k!}$$

$$y = ax + b, \quad \bar{y} = a\bar{x} + b, \quad s_y = |a|\, s_x$$

$$\mu_{\bar{x}} = \mu, \quad \sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}, \quad \mu_{\hat{p}} = p, \quad \sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}}$$

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}, \quad \text{margin of error} = 2 \cdot SE, \quad \bar{x} \pm 2\frac{\sigma}{\sqrt{n}}$$